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Foundational Processes and Growth

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1 Big picture

- **Question.** How do product innovation and two distinct forms of process innovation — *foundational* versus *cost-reducing* — jointly shape firm growth?
- **Core conceptual distinction.** Product innovation creates new varieties; cost-reducing process innovation improves efficiency on the current production frontier; foundational process innovation shifts the technological possibility frontier and unlocks future product creation.
- **Setting.** The paper combines a simple innovation model with patent-text classification using millions of patents from 1900–2020, firm-level evidence for U.S. public firms, cross-country patent-office evidence, and product-level evidence from the FDA Orange Book for pharmaceuticals.
- **Main empirical idea.** Classify patents into product and process patents using titles and claims; then identify *foundational* process patents as those that are dissimilar to a firm’s prior products but similar to its future products.
- **Main result.** Both process and product innovations are positively associated with firm growth, but their timing differs: process innovation matters more in the short run, while product innovation builds gradually. Within process innovation, foundational processes have the strongest and most persistent relation with future growth.
- **Mechanism.** Foundational processes induce more and better future products. Product patents that build on foundational processes have higher market values, more citations, more links to science, and are more likely to expand the firm’s technology space.
- **Economic interpretation.** Foundational processes behave like “deep-tech” corporate innovations. They often rely more on basic science, expand what firms can make, and then generate growth through subsequent product introductions.
- **Why the paper matters.** The paper contributes both a new measurement framework for heterogeneous innovation and a growth model in which process innovation is not just cost saving but also product creating.

2 Data and measurement (institutional setting and key objects)

2.1 Patent sources and overall coverage

- **U.S. patent data.** The paper uses Google Patents for the full history of granted U.S. utility patents, with coverage from 1836 to 2020.
- **International patent data.** The paper uses the EPO–OECD PATSTAT database to classify non-U.S. patents using English patent titles.
- **Scale.** After the title-based classification, the final sample contains more than 50 million classified patents across the U.S. and international offices.
- **Why titles matter.** Claims are richly informative but systematically digitized only for the USPTO post-1976. Titles exist across time and countries, so title-based classification is what makes the international and long-run analysis feasible.

2.2 Classifying process versus product patents

- **Core rule.** Titles and claim preambles that refer to an *activity* (e.g. process, method, use) are classified as process patents; those referring to a *physical entity* (e.g. product, device, apparatus) are classified as product patents.
- **Method.** The authors build an expanding hypernym-based “bag of words” classification rather than relying on a fixed hand-crafted dictionary.
- **Why this is credible.** Patent offices require titles to describe the main subject of the invention, so the title is not cheap talk.
- **Validation.** The title/claim classification overlaps 93.3% with outside expert and patent-examiner classifications (MaxVal Group Inc. and IP Australia). It also correlates strongly with claims-based classifications and with the process-claim approach in Bena and Simintzi (2025).
- **Aggregate shares.** More than 58% of U.S. patents granted from 1900 to 2020 are product patents. Within process patents, about 69% are classified as cost-reducing and the remainder as foundational.

2.3 Classifying foundational versus cost-reducing process patents

- **Key idea.** A foundational process should look *new* relative to a firm’s prior products but should become relevant for the firm’s later products. A cost-reducing process should look more tied to the firm’s current production set.
- **Text similarity engine.** Each patent is represented using TF – IDF vectors, and patent-pair similarity is measured using cosine similarity.
- **Within-firm classification.** For a focal process patent, the paper computes similarity to *past* product patents and to *future* product patents within the same firm.
- **Threshold rule.** Foundational patents are process patents whose within-firm similarity ratio exceeds the yearly 80th percentile among process patents granted to public firms; the rest are treated as cost-reducing.
- **Interpretation.** Foundational patents are processes that open future product possibilities, not just processes that optimize the current production line.

2.4 Validation of the cost-reducing / foundational split

- **Accounting validation.** Cost-reducing patent stock is associated with a lower employee-to-PPE ratio, lower capital expenditures over assets, and lower PPE over assets, consistent with labor saving and higher capital efficiency.
- **Disclosure validation.** Cost-reducing patent stock is positively associated with cost-reduction language in firms’ 10-K/Q filings.
- **Foundational contrast.** Foundational patent stock is instead positively related to capital expenditures, which fits the view that moving to a new production technology often requires new capital investment.

- **Science linkage.** Foundational patents cite substantially more non-patent literature and are more strongly linked to subsequent scientific publications by the firm.

2.5 Firm-level, cross-country, and product-level data

- **Public-firm sample.** The main firm-level evidence uses CRSP–Compustat merged data for U.S. publicly listed firms from 1976 to 2020, combined with patent values from Kogan et al. (2017).
- **Growth outcomes.** The paper studies profits, sales, capital, employment, and revenue-based TFP.
- **Cross-country sample.** Using patent families and the international classification, the authors study how the foundational-process ratio relates to future GDP and TFP growth across countries.
- **Orange Book data.** For product-level evidence, the paper uses the NBER digitized FDA Orange Book. The sample includes 5,511 patents linked to 2,173 distinct small-molecule drug applications from 1985 to 2016.
- **Drug quality outcomes.** Drug value is measured using the stock-market reaction to approval, FDA priority-review status, and health-insurance / Medicaid spending.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Consumer demand and product quality

Consumers live on an outer Salop circle, while the firm’s product varieties sit on an inner ring of radius $s < 1$, which the paper calls the TPF. Demand for a specific variety is

$$c = q(i, j)p^{-\varepsilon}, \quad (1)$$

where $q(i, j)$ is the consumer-product quality match and $\varepsilon > 1$ is the elasticity of substitution across product lines. Quality is

$$q(i, j) = \chi - \lambda(1 - s) - \mu s|i - j|. \quad (2)$$

The first mismatch term, $\lambda(1 - s)$, captures *vertical* quality: if the technological frontier moves outward (higher s), products become intrinsically better. The second term, $\mu s|i - j|$, captures *horizontal* mismatch between a consumer’s taste and the specific variety.

Aggregating across consumers who buy the closest product gives total demand:

$$c = \left[2\pi(\chi - \lambda(1 - s)) - \mu s \frac{\pi^2}{n} \right] p^{-\varepsilon}. \quad (3)$$

- **Interpretation.** Better technology (higher s) improves product quality directly and also raises the value of variety expansion because the firm can now supply better products that more closely match tastes.

- **Key complementarity.** Vertical quality and product variety reinforce each other. This is the paper's central mechanism for why foundational processes stimulate later product innovation.

3.2 Product innovation problem

Given a technological frontier s and production efficiency k (with marginal cost k^{-1}), the firm chooses price p and number of varieties n :

$$\begin{aligned} \max_{p,n} p^{1-\varepsilon} & \left[2\pi(\chi - \lambda(1-s)) - \mu s \frac{\pi^2}{n} \right] \\ & - k^{-1} p^{-\varepsilon} \left[2\pi(\chi - \lambda(1-s)) - \mu s \frac{\pi^2}{n} \right] - \gamma n^{1/\gamma}. \end{aligned} \quad (4)$$

The first-order conditions imply

$$p^* = \frac{\varepsilon}{\varepsilon - 1} k^{-1}, \quad (5)$$

and

$$n^* = \left(\frac{1}{\varepsilon} \left(\frac{\varepsilon}{\varepsilon - 1} k^{-1} \right)^{1-\varepsilon} \mu s \pi^2 \right)^{\gamma/(1+\gamma)}. \quad (6)$$

- **Interpretation of price.** Cost-reducing process innovation works through k : higher efficiency lowers marginal cost and therefore lowers optimal price.
- **Interpretation of variety.** Foundational process innovation works through s : by shifting the frontier outward, it increases the profitability of creating more varieties.
- **Empirical implication.** Products should expand both after cost-reducing and foundational processes, but much more strongly after foundational processes because s raises both quality and the return to variety.

3.3 Process innovation problem

The firm chooses the Poisson arrival rates of two kinds of process innovation:

$$\begin{aligned} \max_{\xi_k, \xi_s} & \xi_k \Pi(k + \Delta k, s) + \xi_s \Pi(k, s + \Delta s) + (1 - \xi_k - \xi_s) \Pi(k, s) \\ & - \eta \xi_k^{1/\eta} - \eta \xi_s^{1/\eta}. \end{aligned} \quad (7)$$

Here ξ_k is the intensity of *cost-reducing* process innovation because it improves efficiency k , while ξ_s is the intensity of *foundational* process innovation because it expands the frontier s .

- **Cost-reducing process innovation.** Raises k , lowers marginal cost, and shifts growth forward mainly through lower prices and short-run efficiency gains.
- **Foundational process innovation.** Raises s , expands the set of feasible high-quality products, and induces future product creation.

- **Notation note.** The model labels the innovation rates by the state variable they move: ξ_k changes k , and ξ_s changes s .

3.4 Growth decomposition in the model

The paper’s reduced-form growth logic can be summarized as

$$\Delta \text{Rev} \approx \Delta k \xi_k \frac{\partial \text{Rev}}{\partial k} + \Delta s \xi_s \frac{\partial \text{Rev}}{\partial s}. \quad (8)$$

The first term is the *cost-reducing* channel: lower marginal cost and lower prices increase revenue. The second term is the *foundational* channel: higher s raises quality directly and also increases variety creation.

The model’s propositions are:

- **Proposition 1.** Foundational process innovation leads to more and higher-quality product introduction.
- **Proposition 2.** Product, foundational process, and cost-reducing process innovations all increase firm growth.
- **Proposition 2a.** The growth contribution of product innovation is largely induced by process innovation, especially foundational process innovation.

3.5 Patent-based measure of foundational processes

For a focal process patent p , the paper defines the similarity ratio

$$q_{p,t}^\tau = \frac{\text{FS}_{p,t}^\tau}{\text{BS}_{p,t}^{-\tau}}, \quad (9)$$

where $\text{FS}_{p,t}^\tau$ is similarity to future product patents within horizon τ , and $\text{BS}_{p,t}^{-\tau}$ is similarity to past product patents within the same horizon. Foundational processes have high forward similarity and low backward similarity.

- **Interpretation.** High- q patents are processes that look unlike the firm’s existing products but become important for its later products.
- **Implementation.** The benchmark horizon is ten years, and foundational patents are the top 20% of process patents by yearly within-firm q .

3.6 Main firm-growth local projection specification

The core reduced-form empirical specification is

$$\ln(Y_{i,t+\tau}) - \ln(Y_{i,t}) = \alpha_0 + \sum_{Pats} \beta_\tau^{Pats} \theta_{i,t}^{Pats} + \gamma_1 \ln(Y_{i,t}) + \gamma_2 \ln(Y_{i,t-1}) + \Gamma X_{i,t} + \alpha_{sic3} + \delta_t + \varepsilon_{i,t}. \quad (10)$$

- **Dependent variables.** Profits, sales, capital, employment, and revenue-based TFP.
- **Innovation regressor.** $\theta_{i,t}^{Pats}$ is the market-value-weighted stock of patents of a given type granted in year t , scaled by total assets.
- **Controls.** Initial outcomes, capital, employment, total assets, and idiosyncratic volatility, with industry and year fixed effects.
- **Interpretation.** The estimated β_{τ}^{Pats} trace out dynamic associations over horizons $\tau = 1, \dots, 10$.

3.7 Cross-country and product-quality regressions

At the country level, the paper estimates

$$\ln(Y_{c,t+\tau}) - \ln(Y_{c,t}) = \beta_{\tau} \frac{Foundational_{c,t}}{Product_{c,t} + CostReducing_{c,t}} + \gamma \ln(Y_{c,t}) + \Lambda X_{c,t} + \alpha_c + \delta_t + \varepsilon_{c,t+\tau}. \quad (11)$$

For product patents, the paper compares product quality outcomes for patents that cite foundational processes:

$$Quality_p = \beta_0 + \beta_1 \mathbb{I}\{\text{cites foundational process}\}_p + \delta_{IPC4 \times year} + \varepsilon_p. \quad (12)$$

For approved drugs, the paper estimates regressions of the form

$$DrugQuality_d = \beta_1 Share_d^{found} + \beta_2 Share_d^{cost} + \beta_3 Share_d^{prod} + \alpha_{f \times t} + \varepsilon_d. \quad (13)$$

- **Interpretation.** The empirical design keeps re-using the same logic: foundational processes should show up in future product quality, product variety, and commercially valuable product introductions.

4 Identification and empirical credibility

4.1 What is and is not identified causally

- **Important distinction.** The main firm-growth results are *not* identified off a clean external natural experiment or an instrument. They are dynamic reduced-form associations guided by the model.
- **What the paper does well.** It uses timing, local projections, rich controls, industry and year fixed effects, and market-value-weighted patent shocks to discipline the analysis.
- **How to read the results.** The paper's strongest contribution is measurement plus theory-consistent empirical patterns, rather than a narrow causal treatment effect from exogenous variation.

4.2 Why the classification exercise is credible

- **Legal-document structure.** Patent titles and claims are unusually suitable for text classification because patent offices require them to describe the invention category clearly.
- **Expert validation.** The 93.3% overlap with MaxVal / IP Australia classifications is strong evidence that the process/product split is not arbitrary.
- **Economic validation.** The cost-reducing split also behaves correctly in the data: it lines up with lower labor intensity, lower capital expenditure intensity, and more cost-cutting language in firm filings.
- **Science validation.** Foundational processes cite more NPL and are more closely linked to later scientific publication output, which is exactly what one would expect from more frontier-shifting innovations.

4.3 Why the foundational q measure is informative

- **Within-firm logic.** The key measure compares a process patent to a firm’s own past and future product patents, which keeps the classification closely tied to internal technology evolution.
- **Interpretation.** High forward similarity plus low backward similarity is exactly the pattern a foundational process should have if it opens new product possibilities.
- **Robustness.** The paper shows similar patterns with alternative q thresholds, claims-based classifications, citation-based definitions, and cross-firm versions of the similarity measure.

4.4 Where the evidence is strongest

- **Patent-quality regressions.** These compare patents within the same technology class and year, so the interpretation is cleaner than raw cross-sectional comparisons.
- **Drug regressions.** The Orange Book evidence includes firm-by-year fixed effects, so it compares drugs within the same firm-time environment based on how much their patent bundles lean on foundational processes.
- **Product-space regressions.** The link between foundational processes and new IPC classes is especially compelling because it maps directly into the model’s “new feasible varieties” mechanism.

4.5 Main caveats

- **Public-firm dependence.** The foundational-versus-cost-reducing split is defined using within-firm similarity for publicly listed firms, so the richest firm-level analysis is not a census of all innovating firms.
- **Drug sample selection.** The FDA analysis only uses approved drugs, so it speaks to the *quality* of successful products, not to approval probabilities.
- **Cross-country regressions.** The country-level growth results are best read as broad external-consistency evidence, not as a clean causal estimate.

5 Tables and figures

5.1 Figure 1 and Figure 2: Consumer preferences, product variety, and the TPF

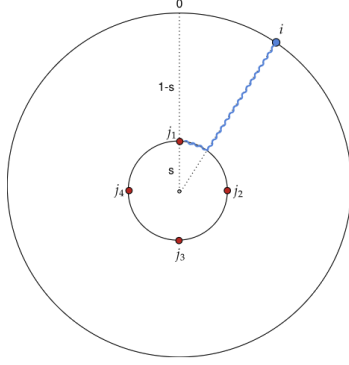
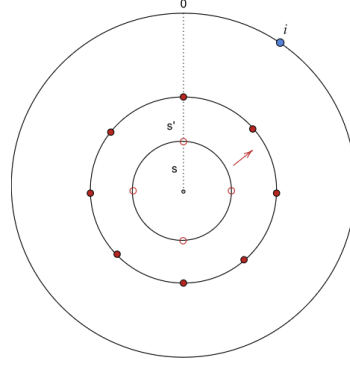


Fig. 1. Consumer preferences and product-consumer match.



Read:

- Figure 1 is the paper's conceptual backbone: consumers sit on the outer Salop circle, while products lie on an inner ring representing the firm's current TPF.
- Figure 2 then shows what a foundational process does: it pushes the inner ring outward from s to s' , raising intrinsic product quality and increasing the return to creating more varieties.
- These figures are crucial because they make the paper's distinction concrete: cost-reducing processes move *along* the current frontier, while foundational processes move the frontier itself.

5.2 Table 1: Validation of cost-reducing patents

Read:

- This is the main validation table for the foundational / cost-reducing split.
- Cost-reducing patent stock is negatively associated with the employee-to-PPE ratio (-0.009), CAPX/AT (-0.059), and PPE/AT (-0.122), which matches the idea that these patents improve efficiency and economize on inputs.
- In 10-K/Q filings, cost-reducing stock is positively associated with cost-cutting language (coefficient 0.016), while product patents are negatively related to that language.
- Foundational patent stock instead raises CAPX/AT (0.086), which fits the idea that moving to a new technological frontier often requires investment rather than simple economizing.

5.3 Table 2: Industry distribution and firm characteristics

Read:

- Product patents dominate in every Fama–French industry, but process-patent shares are still large, typically between 26% and 50%.

Table 1
Validation of cost-reducing patents.
Panel A. Accounting information

	EMP/PPE (1)	CAPX/AT (2)	PPE/AT (3)
Ln(Foundational Stock)	0.001 (3.78)	0.086*** (3.79)	0.051 (3.79)
Ln(Cost-reducing Stock)	-0.009*** (-5.42)	-0.059*** (-3.03)	-0.122*** (-2.29)
Ln(Product Stock)	-0.002 (-0.90)	-0.014 (-0.39)	0.236*** (2.34)
Controls	✓	✓	✓
Year FE	✓	✓	✓
Firm FE	✓	✓	✓
Obs.	148,220	151,774	153,370
Adj. R ²	0.84	0.53	0.87

Panel B. Company filings

	10K/Q (1)
Ln(Foundational)	-0.004 (-0.70)
Ln(Cost-reducing)	0.016*** (2.99)
Ln(Product)	-0.015*** (-3.12)
Controls	✓
Year FE	✓
Firm FE	✓
Obs.	127,519
Adj. R ²	0.45

This table presents validation analysis for the classification of cost-reducing patents. In Panel A, the dependent variables are the number of employees per million of property, plant, and equipment (PPE), capital expenditures (CAPX), and PPE both expressed as shares of total assets. In Panel B, the dependent variable is an indicator variable equal to one when cost-reducing terms are mentioned in firms 10-K/Q filings, and zero otherwise. The search terms include the following and their variants: cost-reducing, reduce cost, operational efficiency, efficiency gain, increase productivity, improve productivity, productivity improvement, process efficiency, cost cutting, reduced labor, operational improvement, overhead reduction, efficiency improvement, cost containment, expense control, workflow optimization, cost control, cost minimization, increase efficiency, improve efficiency, efficiency enhancement, and resource optimization. Foundational and cost-reducing patents are defined using the within-firm q-ratio at the 80th percentile. The included control variables are: Ln(Total assets), Tobin's Q, cash flow, R&D expenditure, and a missing R&D indicator. Standard errors are clustered at the firm level. ***, **, * indicate significance level at 1%, 5% and 10%, respectively. All variable definitions are provided in Table E.2 in the Appendix. The sample period for Panel A is 1976 to 2020, and 1993 to 2020 for Panel B, when 10-K/Q electronic filings are available.

Table 2
Industry distribution and firm characteristics by patents type.

Panel A. Industry distribution	Foundational		Cost-reducing		Process		Product		Total
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	
Oil, Gas, and Coal Extraction and Products	8%	35%	15%	24%	50%	50%	50%	50%	159,484
Finance	15%	24%	26%	26%	49%	50%	50%	50%	31,234
Wholesale, Retail, and Some Services	16%	26%	26%	26%	49%	50%	50%	50%	27,978
Business Equipment	9%	31%	26%	26%	49%	50%	50%	50%	1,198,126
Utilities	2%	6%	26%	26%	49%	50%	50%	50%	3771
Chemicals and Allied Products	6%	26%	26%	26%	49%	50%	50%	50%	201,738
Telephone and Television Transmission	7%	23%	26%	26%	49%	50%	50%	50%	125,909
Other	4%	25%	26%	26%	49%	50%	50%	50%	211,414
Healthcare, Medical Equipment, and Drugs	5%	21%	26%	26%	49%	50%	50%	50%	230,184
Manufacturing	4%	20%	26%	26%	49%	50%	50%	50%	533,659
Consumer Durables	4%	18%	26%	26%	49%	50%	50%	50%	248,519
Consumer NonDurables	4%	12%	26%	26%	49%	50%	50%	50%	37,061

Panel B. Firm Characteristics

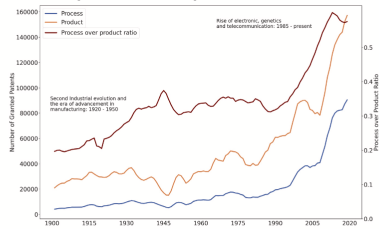
Foundational	Cost-reducing			Product			Non-Foundational		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Ln(Total Asset)	6.93	6.87	2.20	6.55	6.46	2.27	5.98	5.78	2.31
Ln(Market Cap)	6.91	6.86	2.20	6.53	6.45	2.35	5.98	5.83	2.39
Tobin's Q	2.00	1.49	1.50	1.99	1.47	1.53	2.03	1.47	1.65
Cash Holdings	0.19	0.11	0.21	0.19	0.11	0.22	0.20	0.11	0.23
ROA	-0.00	0.05	0.12	-0.02	0.04	0.22	-0.03	0.04	0.24
CAPX	0.05	0.04	0.04	0.05	0.04	0.04	0.05	0.04	0.04
PPE	0.25	0.23	0.17	0.25	0.21	0.17	0.24	0.21	0.17
Sales Growth	0.11	0.07	0.33	0.11	0.07	0.36	0.13	0.08	0.42
R&D	0.07	0.04	0.11	0.12	0.09	0.33	0.16	0.09	0.37
Missing R&D	0.09	0.00	0.20	1.956.54	221.08	5307.03	1.905.31	113.37	4291.96
Profit (\$mil)	2419.47	335.99	6140.44	5.569.20	634.85	15,347.96	3,928.44	324.48	11,734.45
Output (\$mil)	6855.19	953.47	17,772.53	37.12	2.34	123.28	25.58	1.11	89.81
Capital (\$mil)	45.49	3.80	145.04	-0.18	-0.18	0.38	-0.24	-0.23	0.37
TFP	-0.11	-0.13	0.37	38.967	-0.18	0.38	-0.24	-0.23	0.37
N	25,963			58,818			58,818		

The table presents the industry distribution of foundational and cost-reducing process patents, product patents, and firm characteristics for patent-holding firms. Patents are classified based on their titles and the within-firm Q-ratio. Panel A presents the percentage distribution of foundational and cost-reducing process patents, as well as product patents, across each Fama-French twelve-industry classification (sourced from the Ken French Data Library). The sample period spans from 1930 to 2020. Panel B reports characteristics of firms with at least one foundational process patent, cost-reducing process patent, product patent and non-foundational patent. To be included, a firm-year must hold at least one patent in the patent portfolio. A patent is assumed to have a lifespan of twenty years. The dataset comprises a CRSP-Compustat firm-year panel from 1976 to 2020. All variable definitions are provided in Table E.2 in the Appendix.

- Business Equipment is the largest patenting sector in the public-firm sample, and the later figures show that it drives much of the foundational-process surge after 1990.
- Firms with foundational processes are systematically larger and more R&D intensive than firms with no foundational patents, which supports the paper's deep-tech interpretation.

5.4 Figure 3 and Figure 4: Historical evolution of heterogeneous innovation

Panel A. Process and product innovation: all patents



Panel B. Foundational process innovation: publicly listed firms

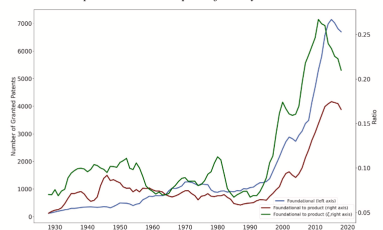


Fig. 3. Historical evolution of heterogeneous innovations.

The figure presents the time-series variation in the number of different types of innovations for U.S. patents granted from 1900 to 2020, as a three-year moving average. Panel A displays the number of process and product patents (left axis) and the process-to-product ratio (right axis) for all USPTO granted patents. Panel B shows the evolution of foundational process patents (left axis) and the foundational process-to-product ratio (right axis) by the number of patents and market value (2) weighted for the subset of U.S. publicly listed firms, available from 1929.

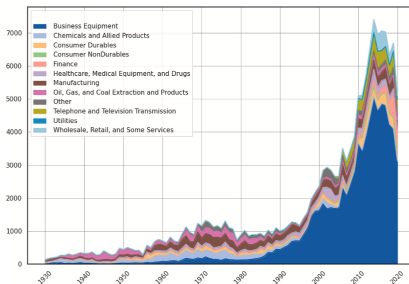


Fig. 4. Foundational process patents across industries.

The figure presents the time-series variation of the number of U.S. foundational patents granted to publicly listed firms by Fama-French 12 industries for the sample period from 1930 to 2020.

Read:

- Figure 3 Panel A shows that process patents rise sharply in two major eras: the post-1920 manufacturing wave and the post-1985 electronics / genetics / telecommunications wave.
- Figure 3 Panel B shows that foundational processes become especially important among public firms after 1990, reaching roughly 17% of product patents by count.

- The market-value-weighted foundational ratio spikes around major technology waves, which is exactly what one would expect if foundational processes are tied to broad technological regime shifts.
- Figure 4 shows the industry composition of foundational patents over time; after 1990, Business Equipment dominates the expansion, consistent with semiconductors, computing, and related production technologies.

5.5 Table 3: Process and foundational patent characteristics

Read:

- Relative to product patents, process patents have higher private value (0.132 in $\ln \xi$), more NPL citations (0.202), more re-assignments (0.066), and higher renewal rates (0.017), even though they are not more forward cited on average.
- The more important comparison is within process patents: foundational patents have 41.6% more forward citations, 32.4% more citations from other firms, 30.2% higher private value, and 16.3% more NPL citations than cost-reducing patents.
- So foundational patents look more socially valuable, more privately valuable, and more science-linked than other process patents.

t.png

Table 5
Firm growth — foundational, cost-reducing process and product patents.

	t+1	t+2	t+3	t+4	t+5	t+6	t+7	t+8	t+9	t+10
Panel A. Profit										
Process, foundational (f)	0.009*** (3.80)	0.010*** (3.82)	0.010*** (3.51)	0.010*** (3.32)	0.010*** (3.20)	0.020*** (2.97)	0.019*** (2.76)	0.018*** (2.51)	0.017*** (2.22)	0.016*** (2.28)
Process, cost-reducing (f)	0.003 (1.00)	0.006 (1.37)	0.014** (2.26)	0.010 (1.37)	0.005 (0.75)	0.006 (0.84)	0.006 (0.69)	0.009 (0.96)	0.011 (1.09)	0.009 (1.09)
Product (f)	0.011*** (3.20)	0.012*** (2.36)	0.007 (0.79)	0.014** (1.96)	0.019** (2.24)	0.020** (2.21)	0.020** (2.01)	0.023** (1.98)	0.022** (1.76)	0.025** (2.18)
Obs.	154,143	138,424	124,709	112,685	102,065	92,644	84,273	76,798	70,063	64,013
Panel B. Sales										
Process, foundational (f)	0.009*** (3.20)	0.010*** (3.71)	0.014*** (3.70)	0.014*** (3.66)	0.017*** (3.97)	0.018*** (3.80)	0.018*** (3.43)	0.016*** (3.09)	0.016*** (2.82)	0.018*** (3.23)
Process, cost-reducing (f)	0.003 (0.93)	0.006 (1.37)	0.008 (1.18)	0.008 (1.32)	0.004 (0.58)	0.002 (0.24)	-0.001 (-0.14)	0.000 (0.02)	0.004 (0.46)	0.001 (0.07)
Product (f)	0.009*** (3.85)	0.012*** (3.16)	0.012*** (2.94)	0.012*** (2.32)	0.019** (1.83)	0.019** (2.16)	0.020** (2.38)	0.023** (2.53)	0.020** (1.83)	0.023** (2.24)
Obs.	154,143	138,424	124,709	112,685	102,065	92,644	84,273	76,798	70,063	64,013
Panel C. Capital										
Process, foundational (f)	0.004*** (2.64)	0.009*** (3.15)	0.013*** (3.38)	0.016*** (3.52)	0.020*** (3.60)	0.022*** (3.70)	0.024*** (3.77)	0.025*** (3.84)	0.025*** (3.91)	0.026*** (3.99)
Process, cost-reducing (f)	0.001 (0.94)	0.003 (1.47)	0.006* (1.66)	0.007 (1.54)	0.005 (1.04)	0.005 (0.59)	-0.001 (-0.16)	-0.002 (-0.29)	-0.000 (-0.03)	-0.000 (-0.03)
Product (f)	0.010*** (3.81)	0.018*** (5.61)	0.022*** (5.19)	0.024*** (4.95)	0.026*** (4.60)	0.026*** (3.75)	0.026*** (3.60)	0.029*** (3.38)	0.027*** (3.12)	0.025*** (2.89)
Obs.	154,143	138,424	124,709	112,685	102,065	92,644	84,273	76,798	70,063	64,013
Panel D. Employment										
Process, foundational (f)	0.003*** (2.66)	0.008*** (3.13)	0.010*** (3.40)	0.011*** (3.47)	0.012*** (3.50)	0.014*** (3.60)	0.014*** (3.71)	0.015*** (3.83)	0.016*** (3.87)	0.016*** (3.90)
Process, cost-reducing (f)	0.002 (1.46)	0.003 (1.47)	0.006* (1.78)	0.006* (1.76)	0.007 (1.75)	0.006 (1.28)	0.004 (0.75)	0.002 (0.35)	0.003 (0.40)	0.003 (0.53)
Product (f)	0.008*** (4.79)	0.013*** (4.54)	0.013*** (3.33)	0.013*** (3.28)	0.012*** (2.42)	0.010** (1.89)	0.012** (1.92)	0.013** (1.94)	0.013** (1.91)	0.012** (1.82)
Obs.	154,143	138,424	124,709	112,685	102,065	92,644	84,273	76,798	70,063	64,013
Panel E. Total Factor Productivity										
Process, foundational (f)	0.004*** (2.83)	0.005*** (2.36)	0.008*** (2.81)	0.008*** (2.92)	0.007*** (2.52)	0.009*** (3.33)	0.009*** (3.42)	0.007*** (2.82)	0.007*** (2.51)	0.007*** (2.79)
Process, cost-reducing (f)	0.004 (1.89)	0.005 (1.42)	0.006* (1.78)	0.007 (1.66)	0.006 (1.33)	0.005 (0.57)	0.005 (0.88)	0.009* (2.26)	0.010** (2.33)	0.011*** (2.64)
Product (f)	0.014*** (4.53)	0.019*** (5.27)	0.025*** (5.55)	0.026*** (5.87)	0.031*** (1.89)	0.031*** (1.83)	0.031*** (1.70)	0.031*** (1.47)	0.031*** (2.03)	0.031*** (2.17)
Obs.	98,859	88,409	79,714	72,084	65,481	59,256	54,412	49,688	45,421	41,640

The table presents relation between foundational, cost-reducing process and product innovation and firm-level outcomes. Patents are classified using titles and within-firm patent similarity. We present the coefficients (β_j) from estimations of the following model:

$$\ln(Y_{i,t+1}) - \ln(Y_{i,t}) = \alpha_0 + \sum_{j=1}^J \beta_j^{(f)} \ln(X_{i,t}^{(f)}) + \beta_1 \ln(X_{i,t}) + \beta_2 \ln(X_{i,t-1}) + \beta_3 \ln(X_{i,t-2}) + \beta_4 \ln(X_{i,t-3}) + \beta_5 \ln(X_{i,t-4}) + \beta_6 \ln(X_{i,t-5}) + \beta_7 \ln(X_{i,t-6}) + \beta_8 \ln(X_{i,t-7}) + \beta_9 \ln(X_{i,t-8}) + \beta_{10} \ln(X_{i,t-9}) + \beta_{11} \ln(X_{i,t-10}) + \beta_{12} \ln(X_{i,t-11}) + \beta_{13} \ln(X_{i,t-12}) + \beta_{14} \ln(X_{i,t-13}) + \beta_{15} \ln(X_{i,t-14}) + \beta_{16} \ln(X_{i,t-15}) + \beta_{17} \ln(X_{i,t-16}) + \beta_{18} \ln(X_{i,t-17}) + \beta_{19} \ln(X_{i,t-18}) + \beta_{20} \ln(X_{i,t-19}) + \beta_{21} \ln(X_{i,t-20}) + \beta_{22} \ln(X_{i,t-21}) + \beta_{23} \ln(X_{i,t-22}) + \beta_{24} \ln(X_{i,t-23}) + \beta_{25} \ln(X_{i,t-24}) + \beta_{26} \ln(X_{i,t-25}) + \beta_{27} \ln(X_{i,t-26}) + \beta_{28} \ln(X_{i,t-27}) + \beta_{29} \ln(X_{i,t-28}) + \beta_{30} \ln(X_{i,t-29}) + \beta_{31} \ln(X_{i,t-30}) + \beta_{32} \ln(X_{i,t-31}) + \beta_{33} \ln(X_{i,t-32}) + \beta_{34} \ln(X_{i,t-33}) + \beta_{35} \ln(X_{i,t-34}) + \beta_{36} \ln(X_{i,t-35}) + \beta_{37} \ln(X_{i,t-36}) + \beta_{38} \ln(X_{i,t-37}) + \beta_{39} \ln(X_{i,t-38}) + \beta_{40} \ln(X_{i,t-39}) + \beta_{41} \ln(X_{i,t-40}) + \beta_{42} \ln(X_{i,t-41}) + \beta_{43} \ln(X_{i,t-42}) + \beta_{44} \ln(X_{i,t-43}) + \beta_{45} \ln(X_{i,t-44}) + \beta_{46} \ln(X_{i,t-45}) + \beta_{47} \ln(X_{i,t-46}) + \beta_{48} \ln(X_{i,t-47}) + \beta_{49} \ln(X_{i,t-48}) + \beta_{50} \ln(X_{i,t-49}) + \beta_{51} \ln(X_{i,t-50}) + \beta_{52} \ln(X_{i,t-51}) + \beta_{53} \ln(X_{i,t-52}) + \beta_{54} \ln(X_{i,t-53}) + \beta_{55} \ln(X_{i,t-54}) + \beta_{56} \ln(X_{i,t-55}) + \beta_{57} \ln(X_{i,t-56}) + \beta_{58} \ln(X_{i,t-57}) + \beta_{59} \ln(X_{i,t-58}) + \beta_{60} \ln(X_{i,t-59}) + \beta_{61} \ln(X_{i,t-60}) + \beta_{62} \ln(X_{i,t-61}) + \beta_{63} \ln(X_{i,t-62}) + \beta_{64} \ln(X_{i,t-63}) + \beta_{65} \ln(X_{i,t-64}) + \beta_{66} \ln(X_{i,t-65}) + \beta_{67} \ln(X_{i,t-66}) + \beta_{68} \ln(X_{i,t-67}) + \beta_{69} \ln(X_{i,t-68}) + \beta_{70} \ln(X_{i,t-69}) + \beta_{71} \ln(X_{i,t-70}) + \beta_{72} \ln(X_{i,t-71}) + \beta_{73} \ln(X_{i,t-72}) + \beta_{74} \ln(X_{i,t-73}) + \beta_{75} \ln(X_{i,t-74}) + \beta_{76} \ln(X_{i,t-75}) + \beta_{77} \ln(X_{i,t-76}) + \beta_{78} \ln(X_{i,t-77}) + \beta_{79} \ln(X_{i,t-78}) + \beta_{80} \ln(X_{i,t-79}) + \beta_{81} \ln(X_{i,t-80}) + \beta_{82} \ln(X_{i,t-81}) + \beta_{83} \ln(X_{i,t-82}) + \beta_{84} \ln(X_{i,t-83}) + \beta_{85} \ln(X_{i,t-84}) + \beta_{86} \ln(X_{i,t-85}) + \beta_{87} \ln(X_{i,t-86}) + \beta_{88} \ln(X_{i,t-87}) + \beta_{89} \ln(X_{i,t-88}) + \beta_{90} \ln(X_{i,t-89}) + \beta_{91} \ln(X_{i,t-90}) + \beta_{92} \ln(X_{i,t-91}) + \beta_{93} \ln(X_{i,t-92}) + \beta_{94} \ln(X_{i,t-93}) + \beta_{95} \ln(X_{i,t-94}) + \beta_{96} \ln(X_{i,t-95}) + \beta_{97} \ln(X_{i,t-96}) + \beta_{98} \ln(X_{i,t-97}) + \beta_{99} \ln(X_{i,t-98}) + \beta_{100} \ln(X_{i,t-99}) + \beta_{101} \ln(X_{i,t-100}) + \beta_{102} \ln(X_{i,t-101}) + \beta_{103} \ln(X_{i,t-102}) + \beta_{104} \ln(X_{i,t-103}) + \beta_{105} \ln(X_{i,t-104}) + \beta_{106} \ln(X_{i,t-105}) + \beta_{107} \ln(X_{i,t-106}) + \beta_{108} \ln(X_{i,t-107}) + \beta_{109} \ln(X_{i,t-108}) + \beta_{110} \ln(X_{i,t-109}) + \beta_{111} \ln(X_{i,t-110}) + \beta_{112} \ln(X_{i,t-111}) + \beta_{113} \ln(X_{i,t-112}) + \beta_{114} \ln(X_{i,t-113}) + \beta_{115} \ln(X_{i,t-114}) + \beta_{116} \ln(X_{i,t-115}) + \beta_{117} \ln(X_{i,t-116}) + \beta_{118} \ln(X_{i,t-117}) + \beta_{119} \ln(X_{i,t-118}) + \beta_{120} \ln(X_{i,t-119}) + \beta_{121} \ln(X_{i,t-120}) + \beta_{122} \ln(X_{i,t-121}) + \beta_{123} \ln(X_{i,t-122}) + \beta_{124} \ln(X_{i,t-123}) + \beta_{125} \ln(X_{i,t-124}) + \beta_{126} \ln(X_{i,t-125}) + \beta_{127} \ln(X_{i,t-126}) + \beta_{128} \ln(X_{i,t-127}) + \beta_{129} \ln(X_{i,t-128}) + \beta_{130} \ln(X_{i,t-129}) + \beta_{131} \ln(X_{i,t-130}) + \beta_{132} \ln(X_{i,t-131}) + \beta_{133} \ln(X_{i,t-132}) + \beta_{134} \ln(X_{i,t-133}) + \beta_{135} \ln(X_{i,t-134}) + \beta_{136} \ln(X_{i,t-135}) + \beta_{137} \ln(X_{i,t-136}) + \beta_{138} \ln(X_{i,t-137}) + \beta_{139} \ln(X_{i,t-138}) + \beta_{140} \ln(X_{i,t-139}) + \beta_{141} \ln(X_{i,t-140}) + \beta_{142} \ln(X_{i,t-141}) + \beta_{143} \ln(X_{i,t-142}) + \beta_{144} \ln(X_{i,t-143}) + \beta_{145} \ln(X_{i,t-144}) + \beta_{146} \ln(X_{i,t-145}) + \beta_{147} \ln(X_{i,t-146}) + \beta_{148} \ln(X_{i,t-147}) + \beta_{149} \ln(X_{i,t-148}) + \beta_{150} \ln(X_{i,t-149}) + \beta_{151} \ln(X_{i,t-150}) + \beta_{152} \ln(X_{i,t-151}) + \beta_{153} \ln(X_{i,t-152}) + \beta_{154} \ln(X_{i,t-153}) + \beta_{155} \ln(X_{i,t-154}) + \beta_{156} \ln(X_{i,t-155}) + \beta_{157} \ln(X_{i,t-156}) + \beta_{158} \ln(X_{i,t-157}) + \beta_{159} \ln(X_{i,t-158}) + \beta_{160} \ln(X_{i,t-159}) + \beta_{161} \ln(X_{i,t-160}) + \beta_{162} \ln(X_{i,t-161}) + \beta_{163} \ln(X_{i,t-162}) + \beta_{164} \ln(X_{i,t-163}) + \beta_{165} \ln(X_{i,t-164}) + \beta_{166} \ln(X_{i,t-165}) + \beta_{167} \ln(X_{i,t-166}) + \beta_{168} \ln(X_{i,t-167}) + \beta_{169} \ln(X_{i,t-168}) + \beta_{170} \ln(X_{i,t-169}) + \beta_{171} \ln(X_{i,t-170}) + \beta_{172} \ln(X_{i,t-171}) + \beta_{173} \ln(X_{i,t-172}) + \beta_{174} \ln(X_{i,t-173}) + \beta_{175} \ln(X_{i,t-174}) + \beta_{176} \ln(X_{i,t-175}) + \beta_{177} \ln(X_{i,t-176}) + \beta_{178} \ln(X_{i,t-177}) + \beta_{179} \ln(X_{i,t-178}) + \beta_{180} \ln(X_{i,t-179}) + \beta_{181} \ln(X_{i,t-180}) + \beta_{182} \ln(X_{i,t-181}) + \beta_{183} \ln(X_{i,t-182}) + \beta_{184} \ln(X_{i,t-183}) + \beta_{185} \ln(X_{i,t-184}) + \beta_{186} \ln(X_{i,t-185}) + \beta_{187} \ln(X_{i,t-186}) + \beta_{188} \ln(X_{i,t-187}) + \beta_{189} \ln(X_{i,t-188}) + \beta_{190} \ln(X_{i,t-189}) + \beta_{191} \ln(X_{i,t-190}) + \beta_{192} \ln(X_{i,t-191}) + \beta_{193} \ln(X_{i,t-192}) + \beta_{194} \ln(X_{i,t-193}) + \beta_{195} \ln(X_{i,t-194}) + \beta_{196} \ln(X_{i,t-195}) + \beta_{197} \ln(X_{i,t-196}) + \beta_{198} \ln(X_{i,t-197}) + \beta_{199} \ln(X_{i,t-198}) + \beta_{200} \ln(X_{i,t-199}) + \beta_{201} \ln(X_{i,t-200}) + \beta_{202} \ln(X_{i,t-201}) + \beta_{203} \ln(X_{i,t-202}) + \beta_{204} \ln(X_{i,t-203}) + \beta_{205} \ln(X_{i,t-204}) + \beta_{206} \ln(X_{i,t-205}) + \beta_{207} \ln(X_{i,t-206}) + \beta_{208} \ln(X_{i,t-207}) + \beta_{209} \ln(X_{i,t-208}) + \beta_{210} \ln(X_{i,t-209}) + \beta_{211} \ln(X_{i,t-210}) + \beta_{212} \ln(X_{i,t-211}) + \beta_{213} \ln(X_{i,t-212}) + \beta_{214} \ln(X_{i,t-213}) + \beta_{215} \ln(X_{i,t-214}) + \beta_{216} \ln(X_{i,t-215}) + \beta_{217} \ln(X_{i,t-216}) + \beta_{218} \ln(X_{i,t-217}) + \beta_{219} \ln(X_{i,t-218}) + \beta_{220} \ln(X_{i,t-219}) + \beta_{221} \ln(X_{i,t-220}) + \beta_{222} \ln(X_{i,t-221}) + \beta_{223} \ln(X_{i,t-222}) + \beta_{224} \ln(X_{i,t-223}) + \beta_{225} \ln(X_{i,t-224}) + \beta_{226} \ln(X_{i,t-225}) + \beta_{227} \ln(X_{i,t-226}) + \beta_{228} \ln(X_{i,t-227}) + \beta_{229} \ln(X_{i,t-228}) + \beta_{230} \ln(X_{i,t-229}) + \beta_{231} \ln(X_{i,t-230}) + \beta_{232} \ln(X_{i,t-231}) + \beta_{233} \ln(X_{i,t-232}) + \beta_{234} \ln(X_{i,t-233}) + \beta_{235} \ln(X_{i,t-234}) + \beta_{236} \ln(X_{i,t-235}) + \beta_{237} \ln(X_{i,t-236}) + \beta_{238} \ln(X_{i,t-237}) + \beta_{239} \ln(X_{i,t-238}) + \beta_{240} \ln(X_{i,t-239}) + \beta_{241} \ln(X_{i,t-240}) + \beta_{242} \ln(X_{i,t-241}) + \beta_{243} \ln(X_{i,t-242}) + \beta_{244} \ln(X_{i,t-243}) + \beta_{245} \ln(X_{i,t-244}) + \beta_{246} \ln(X_{i,t-245}) + \beta_{247} \ln(X_{i,t-246}) + \beta_{248} \ln(X_{i,t-247}) + \beta_{249} \ln(X_{i,t-248}) + \beta_{250} \ln(X_{i,t-249}) + \beta_{251} \ln(X_{i,t-250}) + \beta_{252} \ln(X_{i,t-251}) + \beta_{253} \ln(X_{i,t-252}) + \beta_{254} \ln(X_{i,t-253}) + \beta_{255} \ln(X_{i,t-254}) + \beta_{256} \ln(X_{i,t-255}) + \beta_{257} \ln(X_{i,t-256}) + \beta_{258} \ln(X_{i,t-257}) + \beta_{259} \ln(X_{i,t-258}) + \beta_{260} \ln(X_{i,t-259}) + \beta_{261} \ln(X_{i,t-260}) + \beta_{262} \ln(X_{i,t-261}) + \beta_{263} \ln(X_{i,t-262}) + \beta_{264} \ln(X_{i,t-263}) + \beta_{265} \ln(X_{i,t-264}) + \beta_{266} \ln(X_{i,t-265}) + \beta_{267} \ln(X_{i,t-266}) + \beta_{268} \ln(X_{i,t-267}) + \beta_{269} \ln(X_{i,t-268}) + \beta_{270} \ln(X_{i,t-269}) + \beta_{271} \ln(X_{i,t-270}) + \beta_{272} \ln(X_{i,t-271}) + \beta_{273} \ln(X_{i,t-272}) + \beta_{274} \ln(X_{i,t-273}) + \beta_{275} \ln(X_{i,t-274}) + \beta_{276} \ln(X_{i,t-275}) + \beta_{277} \ln(X_{i,t-276}) + \beta_{278} \ln(X_{i,t-277}) + \beta_{279} \ln(X_{i,t-278}) + \beta_{280} \ln(X_{i,t-279}) + \beta_{281} \ln(X_{i,t-280}) + \beta_{282} \ln(X_{i,t-281}) + \beta_{283} \ln(X_{i,t-282}) + \beta_{284} \ln(X_{i,t-283}) + \beta_{285} \ln(X_{i,t-284}) + \beta_{286} \ln(X_{i,t-285}) + \beta_{287} \ln(X_{i,t-286}) + \beta_{288} \ln(X_{i,t-287}) + \beta_{289} \ln(X_{i,t-288}) + \beta_{290} \ln(X_{i,t-289}) + \beta_{291} \ln(X_{i,t-290}) + \beta_{292} \ln(X_{i,t-291}) + \beta_{293} \ln(X_{i,t-292}) + \beta_{294} \ln(X_{i,t-293}) + \beta_{295} \ln(X_{i,t-294}) + \beta_{296} \ln(X_{i,t-295}) + \beta_{297} \ln(X_{i,t-296}) + \beta_{298} \ln(X_{i,t-297}) + \beta_{299} \ln(X_{i,t-298}) + \beta_{300} \ln(X_{i,t-299}) + \beta_{301} \ln(X_{i,t-300}) + \beta_{302} \ln(X_{i,t-301}) + \beta_{303} \ln(X_{i,t-302}) + \beta_{304} \ln(X_{i,t-303}) + \beta_{305} \ln(X_{i,t-304}) + \beta_{306} \ln(X_{i,t-305}) + \beta_{307} \ln(X_{i,t-306}) + \beta_{308} \ln(X_{i,t-307}) + \beta_{309} \ln(X_{i,t-308}) + \beta_{310} \ln(X_{i,t-309}) + \beta_{311} \ln(X_{i,t-310}) + \beta_{312} \ln(X_{i,t-311}) + \beta_{313} \ln(X_{i,t-312}) + \beta_{314} \ln(X_{i,t-313}) + \beta_{315} \ln(X_{i,t-314}) + \beta_{316} \ln(X_{i,t-315}) + \beta_{317} \ln(X_{i,t-316}) + \beta_{318} \ln(X_{i,t-317}) + \beta_{319} \ln(X_{i,t-318}) + \beta_{320} \ln(X_{i,t-319}) + \beta_{321} \ln(X_{i,t-320}) + \beta_{322} \ln(X_{i,t-321}) + \beta_{323} \ln(X_{i,t-322}) + \beta_{324} \ln(X_{i,t-323}) + \beta_{325} \ln(X_{i,t-324}) + \beta_{326} \ln(X_{i,t-325}) + \beta_{327} \ln(X_{i,t-326}) + \beta_{328} \ln(X_{i,t-327}) + \beta_{329} \ln(X_{i,t-328}) + \beta_{330} \ln(X_{i,t-329}) + \beta_{331} \ln(X_{i,t-330}) + \beta_{332} \ln(X_{i,t-331}) + \beta_{333} \ln(X_{i,t-332}) + \beta_{334} \ln(X_{i,t-333}) + \beta_{335} \ln(X_{i,t-334}) + \beta_{336} \ln(X_{i,t-335}) + \beta_{337} \ln(X_{i,t-336}) + \beta_{338} \ln(X_{i,t-337}) + \beta_{339} \ln(X_{i,t-338}) + \beta_{340} \ln(X_{i,t-339}) + \beta_{341} \ln(X_{i,t-340}) + \beta_{342} \ln(X_{i,t-341}) + \beta_{343} \ln(X_{i,t-342}) + \beta_{344} \ln(X_{i,t-343}) + \beta_{345} \ln(X_{i,t-344}) + \beta_{346} \ln(X_{i,t-345}) + \beta_{347} \ln(X_{i,t-346}) + \beta_{348} \ln(X_{i,t-347}) + \beta_{349} \ln(X_{i,t-348}) + \beta_{350} \ln(X_{i,t-349}) + \beta_{351} \ln(X_{i,t-350}) + \beta_{352} \ln(X_{i,t-351}) + \beta_{353} \ln(X_{i,t-352}) + \beta_{354} \ln(X_{i,t-353}) + \beta_{355} \ln(X_{i,t-354}) + \beta_{356} \ln(X_{i,t-355}) + \beta_{357} 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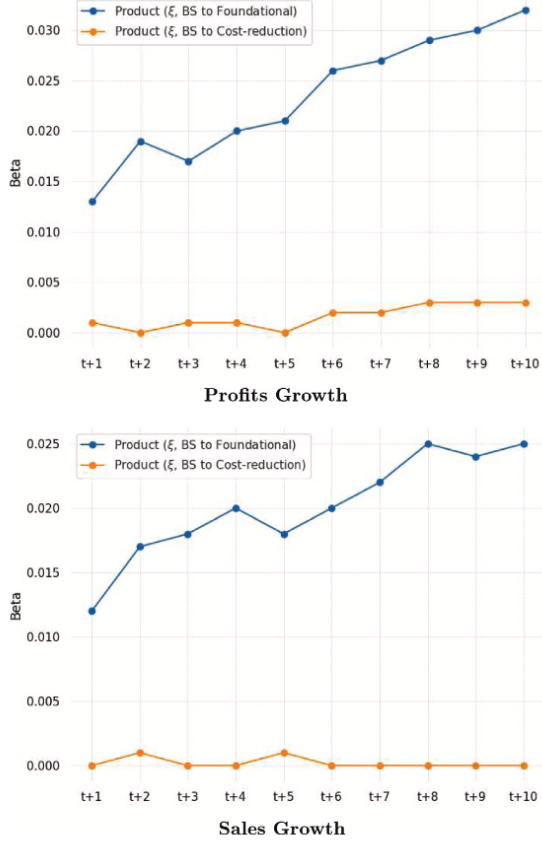


Fig. 7. Growth from product innovations: The role of foundational processes. This figure presents the coefficients from regressions of the relation between firm growth and product innovations that build on foundational vs. cost-reducing patents. We estimate regressions of the form:

$$\ln(Y_{i,t+\tau}) - \ln(Y_{i,t}) = \alpha_0 + \sum_{\tau} \beta_{\tau}^{Pats} \theta_{i,t}^{Pats} + \gamma_1 \ln(Y_{i,t}) + \gamma_2 \ln(Y_{i,t-1}) + \Gamma X'_{i,t} + \alpha_{sic3} + \delta_t + \epsilon_{i,t},$$

where Y represents firm outcomes, $\theta_{i,t}^{Pats}$ is the aggregated market value (ξ) of different types of patents ($Pats$) granted to firm i in year t , with the market value split between foundational (BS to Foundational) and cost-reducing patents (BS to Cost-reducing), scaled by total assets. The values of product patents are weighted according to their backward similarity to foundational and cost-reducing process patents. For example, if a product patent has a market value of \$10 million, with a backward similarity score of 2000 to foundational patents and 3000 to cost-reducing patents, 2/5 or \$4 million would be attributed to foundational patents and \$6 million to cost-reducing patents. $X_{i,t}$ is a vector of control variables, including log capital stock, log employment, log total assets, and idiosyncratic volatility, log process patents, with $\tau \in \{1, \dots, 10\}$. α_{sic3} and δ_t are industry and year fixed effects, respectively. The coefficients for product patents, with their ξ split into foundational process and cost reducing process categories based on the fraction of total backward similarity to these two groups, are plotted. The sample period is 1976 to 2020.

Table 6
Growth from product innovations: The role of foundational processes.

	t+1	t+2	t+3	t+4	t+5	t+6	t+7	t+8	t+9	t+10
Panel A: Profit										
Product (ξ, BS to Foundational)	0.013***	0.019***	0.017***	0.020***	0.021***	0.026***	0.027***	0.029***	0.030***	0.032***
	(4.81)	(3.95)	(2.36)	(3.04)	(2.92)	(3.34)	(3.02)	(2.98)	(2.89)	(3.07)
Product (ξ, BS to Cost-reducing)	0.001	0.000	0.001	0.001	0.000	-0.002	-0.002	-0.003	-0.003	-0.003
	(0.59)	(0.13)	(0.22)	(0.16)	(0.03)	(-0.48)	(-0.54)	(-0.79)	(-0.69)	(-0.77)
Obs.	154,143	138,424	124,709	112,685	102,065	92,644	84,273	76,796	70,063	64,013
Panel B: Sales										
Product (ξ, BS to Foundational)	0.012***	0.017***	0.018***	0.020***	0.018***	0.020***	0.022***	0.025***	0.024***	0.025***
	(6.75)	(5.55)	(4.03)	(3.89)	(2.99)	(3.00)	(3.23)	(3.44)	(2.77)	(2.94)
Product (ξ, BS to Cost-reducing)	-0.000	-0.001	0.000	-0.000	0.001	0.000	0.000	-0.000	-0.000	0.000
	(-0.30)	(-0.36)	(0.08)	(-0.09)	(0.25)	(0.14)	(0.05)	(-0.14)	(-0.04)	(0.02)
Obs.	154,143	138,424	124,709	112,685	102,065	92,644	84,273	76,796	70,063	64,013

The table compares the growth effects of products based on foundational processes to those based on cost-reducing processes. Patents are categorized based on their title, and market values of product patents are weighted according to their backward similarity to both foundational and cost-reducing process patents. For example, if a product patent has a market value of \$10 million, with a backward similarity score of 2000 to foundational patents and 3000 to cost-reducing patents, 2/5 or \$4 million would be attributed to foundational patents and \$6 million to cost-reducing patents. We present the coefficients (β_t) from estimations of the following regression model:

$$\ln(Y_{i,t+\tau}) - \ln(Y_{i,t}) = \alpha_0 + \sum_{\tau} \beta_{\tau}^{Pats} \theta_{i,t}^{Pats} + \gamma_1 \ln(Y_{i,t}) + \gamma_2 \ln(Y_{i,t-1}) + \Gamma X'_{i,t} + \alpha_{sic3} + \delta_t + \epsilon_{i,t},$$

where Y represents firm outcomes, $\theta_{i,t}^{Pats}$ is the aggregated market value (ξ) of different types of product patents ($Pats$) granted to firm i in year t , with the market value split between foundational (BS to Foundational) and cost-reducing patents (BS to Cost-reducing), scaled by total assets. $X_{i,t}$ is a vector of control variables including log capital stock, log of employment, log total asset, and idiosyncratic volatility, and $\tau \in \{1, \dots, 10\}$. α_{sic3} and δ_t are industry and year fixed effects. All right-hand-side variables are scaled to unit standard deviation. Standard errors are clustered at the firm level. All variable definitions are provided in Table B.2 in the Appendix. ***, **, * indicate significance level at 1%, 5% and 10%, respectively. The sample period is 1976 to 2020.

5.9 Table 7: Foundational-process-based product patents

Read:

- Product patents that cite foundational processes are meaningfully better products.
- Relative to other product patents in the same technology-year cell, they have more claims (0.111), more scope (0.030), much more backward scientific grounding (1.073), more originality (0.070), 33.7% more forward citations, 22.5% higher private value, and much more NPL citation intensity (1.246).
- The strongest takeaway is that foundational processes do not just create *more* products; they create higher-value and more science-intensive products.

Table 7
Foundational process-based product patents.

	Claims	Scope	Backward	Originality	Forward	Generality	ln(<i>i</i>)	NPL	Re-assignments	Renewal
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Foundational	0.111*** (16.42)	0.030*** (2.75)	1.073*** (41.64)	0.070*** (25.86)	0.335*** (10.31)	0.025*** (3.48)	0.225*** (12.81)	1.246*** (26.56)	0.001 (0.06)	0.016*** (3.72)
IPC4 × Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Obs.	1,294,575	1,358,484	1,358,482	1,346,312	1,357,453	1,058,761	1,363,490	1,052,982	1,249,618	1,363,490
Adj. R ²			0.21			0.17	0.22			0.39

This table presents patent metrics comparing product patents that cite foundational patents with all other patents. We estimate regressions of the form:
 $Product\ Patent\ Quality_i = \beta_0 + \beta_1 I(Foundational)_i + \beta_{2-10} X_i + \epsilon_i$
 where $Product\ Patent\ Quality_i$ is patent quality metrics, $Foundational_i$ is an indicator variable equal to one for all product patents that cite foundational process patents (own or other firms) and zero otherwise, and $(\beta_2 \times \dots \times \beta_{10})$ are patent class interacted with year fixed effects. All patent metrics in this table are described in detail in 2). Claims is the number of claims by the patent; Scope is the unique number of IPC 4-digit classifications of a patent; Backward is backward citations measured as the number of U.S. patents the patent cites; Originality is the HII index of IPC4 classes of the backward citations; Forward is forward citations measured as number of U.S. patents citing the patent; Generality is the HII index of IPC4 classes of the forward citations; and i is the market value of patent as in *Rogers et al. (2017)*. NPL is the number of non-patent literature citations. Re-assignments is the number of USPTO reported re-assignments of the patent. Renewal is an indicator variable equal to 1 when a patent is renewed after 12 years and zero otherwise. Columns 1, 2, 3, 5, 8 and 9 are estimated using a Poisson specification, while columns 4, 6, 7 and 10 use OLS. T-statistics, adjusted for clustered standard errors at the IPC4 level, are reported in parentheses. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively. All variable definitions are provided in Table E.2 in the Appendix. The sample period is 1970 to 2020, depending on variable information availability.

Table 8
Innovation and product differentiation.
Panel A. Foundational processes and product differentiation

	IPC 3 _{t+1}	IPC 4 _{t+1}	IPC 6 _{t+1}	IPC 7 _{t+1}
	(1)	(2)	(3)	(4)
Ln(Foundational Stock) _t	0.331*** (16.65)	0.346*** (19.08)	0.398*** (17.77)	0.426*** (12.49)
Ln(Cost-reducing Stock) _t	-0.278*** (-16.91)	-0.292*** (-18.50)	-0.282*** (-15.20)	-0.263*** (-11.57)
Ln(Product Stock) _t	-0.061*** (-3.41)	0.010 (0.57)	0.058*** (3.01)	0.090*** (3.66)
R&D	0.840*** (4.81)	0.858*** (5.09)	0.996*** (5.25)	1.155*** (4.68)
Ln(Total Asset)	0.344*** (15.78)	0.345*** (14.70)	0.347*** (12.75)	0.335*** (9.15)
Year FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Obs.	75,261	78,712	81,865	82,932

Panel B. Foundational process contribution to new IPC classes

	New IPC 3	New IPC 4	New IPC 6	New IPC 7
	(1)	(2)	(3)	(4)
BS to Foundational	0.002* (1.88)	0.007*** (3.07)	0.016*** (3.84)	0.013** (2.36)
BS to Cost-reducing	-0.001 (-0.43)	-0.004 (-0.84)	-0.010 (-1.05)	-0.011 (-1.05)
BS to Product	-0.009*** (-3.40)	-0.022*** (-3.62)	-0.050*** (-3.94)	-0.069*** (-4.96)
R&D	-0.168*** (-4.23)	-0.215*** (-3.08)	-0.258** (-2.43)	-0.189* (-1.71)
Ln(Total Asset)	-0.021*** (-6.64)	-0.028*** (-6.43)	-0.034*** (-5.22)	-0.021*** (-2.65)
Year FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Obs.	1,003,543	1,003,543	1,003,543	1,003,543
Adj. R ²	0.18	0.18	0.19	0.18

This table presents results of the analysis of the role of foundational process in expanding the firm's product space. Panel A presents results of regressions of the number of new technology classes measured by 3-digit IPC main classes, 4-, 6-, and 7-digit IPC subclasses at $t+1$ and the stock of different types of patents at time t . The explanatory variables are the natural log of the stock of foundational process, cost-reducing, and product patents up to time t . The sample includes firms that have at least one patent in their portfolio, where a patent has a maximum life span of 20 years. Panel B presents analysis at the product patent level. The dependent variable is an indicator equal to one if a patent belongs to a technology class new to firm i , and zero otherwise. All BS variables measure the backward similarity of the focal patent p to prior foundational, cost-reducing, and other product patents; they are standardized to unit standard deviation. Standard errors are clustered at the firm level. ***, **, and * indicate significance level at 1%, 5% and 10%, respectively. All variable definitions are provided in Table E3 in the Appendix. The sample period is 1976 to 2020.

5.10 Table 8: Innovation and product differentiation

Read:

- Panel A shows that foundational patent stock expands the firm's future technology space. The strongest coefficient is for IPC7 classes: 0.426, implying that a 10% increase in foundational stock is associated with roughly 4.3% more fine-grained technology classes next period.
- Cost-reducing patent stock is negative in all four IPC-count regressions, consistent with these processes improving the current production line rather than opening new lines.
- Panel B shows the same idea at the patent level: patents with higher backward similarity to foundational processes are more likely to enter new IPC classes, while similarity to prior product patents is negative.
- This is the paper's cleanest evidence that foundational processes expand the firm's product opportunity set.

Table 9
Foundational processes and FDA drug quality.

	ln(Abnormal Returns)	Priority	Ln(Patient Spending)
	(1)	(2)	(3)
Foundational (Fraction of Orange Book)	0.523*** (2.64)	0.497*** (3.60)	2.310*** (12.66)
Cost-reducing (Fraction of Orange Book)	-0.084 (-0.35)	0.162*** (6.28)	-0.387 (-0.60)
Product (Fraction of Orange Book)	-0.059 (-0.93)	0.141** (2.30)	0.111 (0.31)
Firm $\times$ Year FE	✓	✓	✓
Obs.	632	632	261
Adj. R^2	0.94	0.68	0.78

This table presents the results on the relation between drug product quality and foundational patents. The dataset consists of unique drug products approved and listed in the Orange Book. We examine three key variables: *Abnormal returns* calculated using the methodology of *Kogan et al. (2017)*, which captures the 3-day market reaction to drug approvals; *Priority* an indicator variable equal to one if the drug is under Priority Review by the FDA and zero otherwise; and *Patient Spending* is drug spending in health insurance programs (Part B, Part D, and Medicaid) from 2017 to 2021. Priority Review is reserved for drugs that offer significant advancements in treatment, prevention, or diagnosis of diseases, lack alternative treatments, provide distinct advantages over existing treatments, or address public health emergencies or national health issues. The explanatory variables are the fraction of foundational, cost-reducing and product patents listed in the Orange Book. Standard errors are clustered at the firm level. ***, **, * indicate significance level at 1%, 5% and 10%, respectively. All variable definitions are provided in Table E.2 in the Appendix. The sample period is 1982 to 2020.

5.11 Table 9: Foundational processes and FDA drug quality

Read:

- The Orange Book results move beyond patents and connect foundational processes to actual product success.
- Drugs whose patent bundles lean more heavily on foundational processes have larger approval-day stock-market reactions (coefficient 0.523), are more likely to receive priority review (0.497), and have much higher patient spending (2.310 in log terms).
- Cost-reducing and generic product shares do not generate similarly strong positive patterns.
- This table is powerful because it shows that foundational processes are not just linked to better patents; they are linked to more valuable real products.

6 Short summary

- **Conclusion.** Foundational process innovation is the process margin that matters most for sustained firm growth because it enables later product creation rather than merely lowering current costs.
- **Intuition.** Cost-reducing processes help firms make the same things more cheaply; foundational processes change what firms are able to make.
- **Empirical lesson.** Patent text can be used to separate process from product innovation and, within process innovation, to isolate a frontier-shifting subset with strong links to future growth, science, and product quality.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper establishes that process innovation is heterogeneous and that foundational processes, not just cost-reducing processes, are central to firm growth.
- **Timing result.** Process innovation has a stronger short-run relation with growth, while product innovation unfolds more gradually; but the persistent product effect is largely concentrated in products enabled by foundational processes.

- **Measurement contribution.** The authors provide a scalable and validated patent-text framework to classify product, cost-reducing process, and foundational process innovation over a very long horizon and across countries.
- **Mechanism result.** Foundational processes induce more and better products: they raise product quality, expand product variety, enlarge the technology space, and improve the commercial success of downstream drugs.
- **Science linkage.** Foundational processes look more like science-based, deep-tech innovation. They cite more NPL, are more linked to later scientific publication, and generate more spillovers.
- **Scope note.** The paper’s main growth evidence is best read as disciplined reduced-form association rather than clean causal treatment identification, but the patterns are coherent across theory, validation exercises, firm outcomes, product outcomes, and macro evidence.

7.2 Economic intuition

- A firm can grow by making current production cheaper or by discovering a new way of producing that makes new products feasible. The paper argues that these are economically distinct margins and should not be collapsed into one “innovation” measure.
- Foundational processes shift the technological frontier outward. Once that happens, the return to product variety creation rises because the new products can be both better and more differentiated.
- That is why foundational processes show up first as science-heavy patents, then as better product patents, and finally as sustained revenue and sales growth.
- Cost-reducing processes still matter, but mostly through short-run efficiency rather than through a persistent expansion of the firm’s opportunity set.

7.3 Takeaway

This paper’s central message is that not all process innovation is about doing the same thing more cheaply. Some processes are *foundational*: they change the firm’s production technology in a way that creates new downstream product possibilities. Once innovation is measured this way, the paper shows a coherent chain from foundational-process discovery to better products to sustained firm growth, making a strong case that frontier-shifting process innovation deserves a central place in growth and corporate-innovation research.